# Oliver H. E. Philcox, Ph.D.

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## CURRENT POSITION

**Department of Physics, Stanford University, Stanford, USA** *2025 - Present*
Assistant Professor

Leinweber Institute for Theoretical Physics at Stanford, *Faculty*
Kavli Institute for Particle Astrophysics and Cosmology, *Senior Member*

## PREVIOUS POSITIONS

**Simons Society of Fellows, New York, USA** *2022 - 2025*
Junior Fellow

**Department of Physics, Columbia University, New York, USA** *2022 - 2025*
Postdoctoral Research Scientist

*Mentors:* Prof. J. Colin Hill & Prof. Lam Hui

## EDUCATION

**Department of Astrophysical Sciences, Princeton University, USA** *2019 - 2022*
Ph.D. in Astrophysics (2022)

**Thesis**: 'Probing Fundamental Cosmology with Galaxy Surveys'

*Thesis Advisors:* David N. Spergel & Matias Zaldarriaga

M.A. in Astrophysics (2020)

**Center for Astrophysics | Harvard & Smithsonian, Cambridge, USA** *2018 - 2019*
Pre-Doctoral Student, *Herchel-Smith Scholar*

*Advisor:* Daniel J. Eisenstein

**Institute of Astronomy, University of Cambridge, UK** *2017 - 2018*
M.Sci. in Astrophysics

**Part III:** 1st Class (Rank 1/28, 97%)

**Thesis:** 'Detection and Removal of B-mode CMB Dust Foregrounds with Signatures of Statistical Anisotropy'

*Thesis Advisor*: Blake D. Sherwin

**Emmanuel College, University of Cambridge, UK** *2014 - 2017*
M.A. (Cantab.) in Natural Sciences, *Senior Scholar*

**Parts IA, IB, II:** 1st Class (Rank 1/20, 90%)

## LONG-TERM ACADEMIC VISITS

**Center for Computational Astrophysics** Jul. 2021 - Aug. 2025
*Guest Researcher* *New York, USA*

**Institute for Advanced Study** Sep. 2020 - Jul. 2022
*Visiting Graduate Student with Matias Zaldarriaga* *Princeton, USA*

**Max-Planck Institute for Astrophysics** Aug. - Sep. 2020
*Visiting Graduate Student with Eiichiro Komatsu* *Munich, Germany*

**Department of Applied Mathematics and Theoretical Physics** May - Jul. 2020
*Visiting Graduate Student with Blake D. Sherwin* *Cambridge, UK*

**Max-Planck-Institut für Astronomie** Jul. - Sep. 2017
*Summer Intern with Jan Rybizki* *Heidelberg, Germany*

**Center for Astrophysics | Harvard & Smithsonian** Jun. - Aug. 2016
*Undergraduate Research Fellow with Ákos Bogdán* *Cambridge, USA*

## PUBLICATION LIST

* = *Author list alphabetized*

### *Major Author*

1. *Chen, S.-F., DeRose, J., Ivanov, M. M., **Philcox, O. H. E.**, "Cosmic Shear in Effective Field Theory at Two-Loop Order: Revisiting $S_8$ in Dark Energy Survey Data", (arXiv).
2. **Philcox, O. H. E.**, "What Shape is the Inflationary Bispectrum?", (arXiv).
3. ***Philcox, O. H. E.**, Pimentel, G. L., Yang, C., "Searching for Unparticles with the Cosmic Microwave Background", (arXiv).
4. *Chudaykin, A., Ivanov, M. M., **Philcox, O. H. E.**, "Reanalyzing DESI DR1: 5. Cosmological Constraints with Simulation-Based Priors", (arXiv).
5. Ivanov, M. M., Sullivan, J. M., Chen, S.-F., Chudaykin, A., Maus, M., **Philcox, O. H. E.**, "Reanalyzing DESI DR1: 4. Percent-Level Cosmological Constraints from Combined Probes and Robust Evidence for the Normal Neutrino Mass Hierarchy", *submitted to Phys. Rev. D* (arXiv).
6. *Chudaykin, A., Ivanov, M. M., **Philcox, O. H. E.**, "Reanalyzing DESI DR1. III. Constraints on inflation from galaxy power spectra and bispectra", *Phys. Rev. D* **113**, 063552 (2026) (arXiv).
7. *Chudaykin, A., Ivanov, M. M., **Philcox, O. H. E.**, "Reanalyzing DESI DR1: 2. Constraints on Dark Energy, Spatial Curvature, and Neutrino Masses", *submitted to Phys. Rev. D* (arXiv).
8. **Philcox, O. H. E.**, Zhong, K., Sirletti, S. S., "Separating the Inseparable: Constraining Arbitrary Primordial Bispectra with Cosmic Microwave Background Data", *submitted to Phys. Rev. D* (arXiv).
9. El-Haj, L. H. A., **Philcox, O. H. E.**, Hill, J. C., "Constraining Inflationary Particle Production with CMB Polarization", *submitted to Phys. Rev. D* (arXiv).
10. *Bakx, T., Ivanov, M. M., **Philcox, O. H. E.**, Vlah, Z., "One-Loop Galaxy Bispectrum: Consistent Theory, Efficient Analysis with COBRA, and Implications for Cosmological Parameters", *submitted to Phys. Rev. D* (arXiv).
11. *Chudaykin, A., Ivanov, M. M., **Philcox, O. H. E.**, "Reanalyzing DESI DR1. I. ΛCDM constraints from the power spectrum and bispectrum", *Phys. Rev. D* **113**, 063502 (2026) (arXiv).
12. ***Philcox, O. H. E.**, Silverstein, E., Torroba, G., "Quantum stress-energy at timelike boundaries: Testing a new beyond-ΛCDM parameter with cosmological data", *Phys. Rev. D* **113**, 043548 (2026) (arXiv).
13. **Philcox, O. H. E.**, Hill, J. C., "Integrated Sachs-Wolfe lensing bispectrum and trispectrum", *Phys. Rev. D* **111**, 123554 (2025) (arXiv).
14. Goldstein, S., **Philcox, O. H. E.**, Fondi, E., Coulton, W. R., "Wonderings on wiggly bispectra: Nonlinear evolution and reconstruction of oscillations in the squeezed bispectrum", *Phys. Rev. D* **112**, 083503 (2025) (arXiv).
15. Zhang, G., Modi, C., **Philcox, O. H. E.**, "Modeling galaxy surveys with hybrid simulation-based inference", *Phys. Rev. D* **113**, 043518 (2026) (arXiv).
16. **Philcox, O. H. E.**, "Searching for inflationary physics with the CMB trispectrum. III. Constraints from *Planck*", *Phys. Rev. D* **111**, 123534 (2025) (arXiv).
17. **Philcox, O. H. E.**, "Searching for inflationary physics with the CMB trispectrum. II. Code and validation", *Phys. Rev. D* **111**, 123533 (2025) (arXiv).

18. **Philcox, O. H. E.**, "Searching for inflationary physics with the CMB trispectrum. I. Primordial theory and optimal estimators", *Phys. Rev. D* **111**, 123532 (2025) (arXiv).

19. ***Philcox, O. H. E.**, Shiraishi, M., "Non-Gaussianity beyond the scalar sector: A search for tensor and mixed tensor-scalar bispectra with Planck data", *Phys. Rev. D* **111**, 123502 (2025) (arXiv).

20. Goldstein, S., **Philcox, O. H. E.**, Hill, J. C., Hui, L., "Intermediate mass-range particles from small scales: Nonperturbative techniques for cosmological collider physics from large-scale structure surveys", *Phys. Rev. D* **110**, 083516 (2024) (arXiv).

21. *Coulton, W. R., **Philcox, O. H. E.**, Villaescusa-Navarro, F., "The Impact of Non-Gaussian Primordial Tails on Cosmological Observables", *submitted to Phys. Rev. D* (arXiv).

22. *Chen, S.-F., Ivanov, M. M., **Philcox, O. H. E.**, Wenzl, L., "Suppression without Thawing: Constraining Structure Formation and Dark Energy with Galaxy Clustering", *Phys. Rev. Lett.* **133**, 231001 (2024) (arXiv).

23. **Philcox, O. H. E.**, Kumar, S., Hill, J. C., "Searching for inflationary particle production in Planck data", *Phys. Rev. D* **111**, 103523 (2025) (arXiv).

24. **Philcox, O. H. E.**, Flöss, T., "Suite of optimal and efficient power spectrum and bispectrum estimators for large-scale structure analyses", *Phys. Rev. D* **112**, 063507 (2025) (arXiv).

25. Cabass, G., **Philcox, O. H. E.**, Ivanov, M. M., Akitsu, K., Chen, S.-F., Simonović, M., Zaldarriaga, M., "BOSS constraints on massive particles during inflation: The cosmological collider in action", *Phys. Rev. D* **111**, 063510 (2025) (arXiv).

26. de Belsunce, R., **Philcox, O. H. E.**, Irsic, V., McDonald, P., Guy, J., Palanque-Delabrouille, N., "The 3D Lyman-$\alpha$ forest power spectrum from eBOSS DR16", *MNRAS* **533**, 3756 – 3770 (2024) (arXiv).

27. **Philcox, O. H. E.**, Ereza, J., "Could Sample Variance be Responsible for the Parity-Violating Signal Seen in the BOSS Galaxy Survey?", *Phil. Trans. Roy. Soc. A* **383**, 2290 (2025) (arXiv).

28. ***Philcox, O. H. E.**, Shiraishi, M., "Testing graviton parity and Gaussianity with Planck $T$-, $E$-, and $B$-mode bispectra", *Phys. Rev. D* **109**, 063522 (2024) (arXiv).

29. Goldstein, S., **Philcox, O. H. E.**, Hill, J. C., Esposito, A., Hui, L., "Consistently constraining $f_{\rm NL}$ with the squeezed lensing bispectrum using consistency relations", *Phys. Rev. D* **109**, 043515 (2024) (arXiv).

30. *Modi, C., **Philcox, O. H. E.**, "Hybrid SBI or How I Learned to Stop Worrying and Learn the Likelihood", *submitted to Phys. Rev. Lett.* (arXiv).

31. **Philcox, O. H. E.**, König, M. J., Alexander, S., Spergel, D. N., "What can galaxy shapes tell us about physics beyond the standard model?", *Phys. Rev. D* **109**, 063541 (2024) (arXiv).

32. ***Philcox, O. H. E.**, Shiraishi, M., "Testing parity symmetry with the polarized cosmic microwave background", *Phys. Rev. D* **109**, 083514 (2024) (arXiv).

33. *Coulton, W. R., **Philcox, O. H. E.**, Villaescusa-Navarro, F., "Signatures of a parity-violating universe", *Phys. Rev. D* **109**, 023531 (2024) (arXiv).

34. **Philcox, O. H. E.**, "Optimal estimation of the binned mask-free power spectrum, bispectrum, and trispectrum on the full sky: Tensor edition", *Phys. Rev. D* **108**, 063506 (2023) (arXiv).

35. *Ivanov, M. M., **Philcox, O. H. E.**, "Measuring $H_0$ with Spectroscopic Surveys", *chapter in "Hubble Constant Tension" (Eds. Di Valentino, E. and Brout, D., Springer, Singapore)* (arXiv).

36. **Philcox, O. H. E.**, "Do the CMB Temperature Fluctuations Conserve Parity?", *Phys. Rev. Lett.* **131**, 181001 (2023) (arXiv).

37. **Philcox, O. H. E.**, "Optimal estimation of the binned mask-free power spectrum, bispectrum, and trispectrum on the full sky: Scalar edition", *Phys. Rev. D* **107**, 123516 (2023) (arXiv).

38. Creque-Sarbinowski, C., Alexander, S., Kamionkowski, M., **Philcox, O. H. E.**, "Parity-violating trispectrum from Chern-Simons gravity", *JCAP* 11 029 (2023) (arXiv).

39. Surrao, K. M., **Philcox, O. H. E.**, Hill, J. C., "Accurate estimation of angular power spectra for maps with correlated masks", *Phys. Rev. D* **107**, 083521 (2023) (arXiv).

40. Ivanov, M. M., **Philcox, O. H. E.**, Cabass, G., Nishimichi, T., Simonović, M., Zaldarriaga, M., "Cosmology with the galaxy bispectrum multipoles: Optimal estimation and application to BOSS data", *Phys. Rev. D* **107**, 083515 (2023) (arXiv).

41. *Cabass, G., Ivanov, M. M., **Philcox, O. H. E.**, Simonovic, M., Zaldarriaga, M., "Constraining single-field inflation with MegaMapper", *Phys. Lett. B* **841**, 137912 (2023) (arXiv).

42. *Cabass, G., Ivanov, M. M., **Philcox, O. H. E.**, "Colliders and ghosts: Constraining inflation with the parity-odd galaxy four-point function", *Phys. Rev. D* **107**, 023523 (2023) (arXiv).

43. Goldstein, S., Esposito, A., **Philcox, O. H. E.**, Hui, L., Hill, J. C., Scoccimarro, R., Abitbol, M. H., "Squeezing $f_{\rm NL}$ out of the matter bispectrum with consistency relations", *Phys. Rev. D* **106**, 123525 (2022) (arXiv).

44. ***Philcox, O. H. E.**, Torquato, S., "Disordered Heterogeneous Universe: Galaxy Distribution and Clustering across Length Scales", *Phys. Rev. X* **13**, 011038 (2023) (arXiv).

45. **Philcox, O. H. E.**, Johnson, M. C., "Novel cosmological tests from combining galaxy lensing and the polarized Sunyaev-Zel'dovich effect", *Phys. Rev. D* **106**, 083501 (2022) (arXiv).

46. **Philcox, O. H. E.**, "Probing parity violation with the four-point correlation function of BOSS galaxies", *Phys. Rev. D* **106**, 063501 (2022) (arXiv).

47. **Philcox, O. H. E.**, Ivanov, M. M., Cabass, G., Simonović, M., Zaldarriaga, M., Nishimichi, T., "Cosmology with the redshift-space galaxy bispectrum monopole at one-loop order", *Phys. Rev. D* **106**, 043530 (2022) (arXiv).

48. **Philcox, O. H. E.**, Farren, G. S., Sherwin, B. D., Baxter, E. J., Brout, D. J., "Determining the Hubble constant without the sound horizon: A 3.6% constraint on $H_0$ from galaxy surveys, CMB lensing, and supernovae", *Phys. Rev. D* **106**, 063530 (2022) (arXiv).

49. *Cabass, G., Ivanov, M. M., **Philcox, O. H. E.**, Simonović, M., Zaldarriaga, M., "Constraints on multifield inflation from the BOSS galaxy survey", *Phys. Rev. D* **106**, 043506 (2022) (arXiv).

50. *Cabass, G., Ivanov, M. M., **Philcox, O. H. E.**, Simonović, M., Zaldarriaga, M., "Constraints on Single-Field Inflation from the BOSS Galaxy Survey", *Phys. Rev. Lett.* **129**, 021301 (2022) (arXiv).

51. *Farren, G. S., **Philcox, O. H. E.**, Sherwin, B. D., "Determining the Hubble constant without the sound horizon: Perspectives with future galaxy surveys", *Phys. Rev. D* **105**, 063503 (2022) (arXiv).

52. **Philcox, O. H. E.**, Ivanov, M. M., "BOSS DR12 full-shape cosmology: ΛCDM constraints from the large-scale galaxy power spectrum and bispectrum monopole", *Phys. Rev. D* **105**, 043517 (2022) (arXiv).

53. Ivanov, M. M., **Philcox, O. H. E.**, Nishimichi, T., Simonović, M., Takada, M., Zaldarriaga, M., "Precision analysis of the redshift-space galaxy bispectrum", *Phys. Rev. D* **105**, 063512 (2022) (arXiv).

54. Ivanov, M. M., **Philcox, O. H. E.**, Simonović, M., Zaldarriaga, M., Nischimichi, T., Takada, M., "Cosmological constraints without nonlinear redshift-space distortions", *Phys. Rev. D* **105**, 043531 (2022) (arXiv).

55. **Philcox, O. H. E.**, Hou, J., Slepian, Z., "A First Detection of the Connected 4-Point Correlation Function of Galaxies Using the BOSS CMASS Sample", *submitted to Phys. Rev. D* (arXiv).

56. **Philcox, O. H. E.**, "Cosmology without window functions. II. Cubic estimators for the galaxy bispectrum", *Phys. Rev. D* **104**, 123529 (2021) (arXiv).

57. ***Philcox, O. H. E.**, Slepian, Z., "Efficient computation of N-point correlation functions in D dimensions", *PNAS* **119**, 33 (2022) (arXiv).

58. **Philcox, O. H. E.**, Slepian, Z., Hou, J., Warner, C., Cahn, R. N., Eisenstein, D. J., "encore: an O (Ng2) estimator for galaxy N-point correlation functions", *MNRAS* **509**, 2457 – 2481 (2021) (arXiv).

59. ***Philcox, O. H. E.**, Slepian, Z., "An exact integral-to-sum relation for products of Bessel functions", *Proc. Roy. Soc. A* **477**, 2253 (2021) (arXiv).

60. Slepian, Z., **Philcox, O. H. E.**, "A uniform spherical goat (problem): explicit solution for homologous collapse's radial evolution in time", *MNRAS* **522**, L42-L45 (2023) (arXiv).

61. ***Philcox, O. H. E.**, Slepian, Z., "Beyond the Yamamoto approximation: Anisotropic power spectra and correlation functions with pairwise lines of sight", *Phys. Rev. D* **103**, 123509 (2021) (arXiv).

62. **Philcox, O. H. E.**, "Cosmology without window functions: Quadratic estimators for the galaxy power spectrum", *Phys. Rev. D* **103**, 103504 (2021) (arXiv).

63. **Philcox, O. H. E.**, Aviles, A., Massara, E., "Modeling the Marked Spectrum of Matter and Biased Tracers in Real- and Redshift-Space", *JCAP* 03 038 (2021) (arXiv).

64. **Philcox, O. H. E.**, Ivanov, M. M., Zaldarriaga, M., Simonovic, M., Schmittfull, M., "Fewer Mocks and Less Noise: Reducing the Dimensionality of Cosmological Observables with Subspace Projections", *Phys. Rev. D* **103**, 043508 (2021) (arXiv).

65. **Philcox, O. H. E.**, Sherwin, B. D., Farren, G. S., Baxter, E. J., "Determining the Hubble Constant without the Sound Horizon: Measurements from Galaxy Surveys", *Phys. Rev. D* **103**, 023538 (2021) (arXiv).

66. **Philcox, O. H. E.**, Massara, E., Spergel, D. N., "What does the marked power spectrum measure? Insights from perturbation theory", *Phys. Rev. D* **102**, 043516 (2020) (arXiv).

67. **Philcox, O. H. E.**, "A faster Fourier transform? Computing small-scale power spectra and bispectra for cosmological simulations in $\mathcal{O}(N^2)$ time", *MNRAS* **501**, 4004 – 4034 (2021) (arXiv).

68. ***Philcox, O. H. E.**, Spergel, D. N., Villaescusa-Navarro, F., "Effective halo model: Creating a physical and accurate model of the matter power spectrum and cluster counts", *Phys. Rev. D* **101**, 123520 (2020) (arXiv).

69. **Philcox, O. H. E.**, Ivanov, M. M., Simonović, M., Zaldarriaga, M., "Combining Full-Shape and BAO Analyses of Galaxy Power Spectra: A 1.6% CMB-independent constraint on $H_0$", *JCAP* 05 032 (2020) (arXiv).

70. **Philcox, O. H. E.**, Eisenstein, D. J., "Computing the Small-Scale Galaxy Power Spectrum and Bispectrum in Configuration-Space", *MNRAS* **492**, 1214 – 1242 (2020) (arXiv).

71. **Philcox, O. H. E.**, Eisenstein, D. J., "Estimating Covariance Matrices for Two- and Three-Point Correlation Function Moments in Arbitrary Survey Geometries", *MNRAS* **490**, 5931 – 5951 (2019) (arXiv).

72. ***Philcox, O. H. E.**, Rybizki, J., "Inferring Galactic Parameters from Chemical Abundances: A Multi-Star Approach", *ApJ* **887**, 9 (2019) (arXiv).

73. **Philcox, O. H. E.**, Eisenstein, D. J., O'Connell, R., Wiegand, A., "rascalc: a jackknife approach to estimating single- and multitracer galaxy covariance matrices", *MNRAS* **491**, 3290 – 3317 (2020) (arXiv).

74. ***Philcox, O. H. E.**, Sherwin, B. D., van Engelen, A., "Detection and Removal of B-mode Dust Foregrounds with Signatures of Statistical Anisotropy", *MNRAS* **479**, 5577 – 5595 (2018) (arXiv).

75. **Philcox, O. H. E.**, Rybizki, J., Gutcke, T., "On the Optimal Choice of Nucleosynthetic Yields, Initial Mass Function, and Number of SNe Ia for Chemical Evolution Modeling", *ApJ* **861**, 40 (2018) (arXiv).

***Contributing Author***

76. *Besuner, R., *et al.* (inc. **Philcox, O. H. E.**), "The Spectroscopic Stage-5 Experiment", *White Paper* (arXiv).

77. *Krause, E., *et al.* (inc. **Philcox, O. H. E.**), "A Parameter-Masked Mock Data Challenge for Beyond-Two-Point Galaxy Clustering Statistics", *ApJ* **990**, 99 (2025) (arXiv).

78. Rogers, K. K., Hložek, R., Laguë, A., Ivanov, M. M., **Philcox, O. H. E.**, *et al.*, "Ultra-light axions and the $S_8$ tension: joint constraints from the cosmic microwave background and galaxy clustering", *JCAP* 06 023 (2023) (arXiv).

79. *Abdalla, E., *et al.* (inc. **Philcox, O. H. E.**), "Cosmology intertwined: A review of the particle physics, astrophysics, and cosmology associated with the cosmological tensions and anomalies", *JHEA* **34**, 49 – 211 (2022) (arXiv).

80. Villaescusa-Navarro, F., *et al.* (inc. **Philcox, O. H. E.**), "The CAMELS Project: Public Data Release", *ApJS* **265**, 54 (2023) (arXiv).

81. Hou, J., Cahn, R. N., **Philcox, O. H. E.**, Slepian, Z., "Analytic Gaussian covariance matrices for galaxy $N$-point correlation functions", *Phys. Rev. D* **106**, 043515 (2022) (arXiv).

82. Schmittfull, M., Simonović, M., Ivanov, M. M., **Philcox, O. H. E.**, Zaldarriaga, M., "Modeling Galaxies in Redshift Space at the Field Level", *JCAP* 05 059 (2021) (arXiv).

83. Villaescusa-Navarro, F., *et al.* (inc. **Philcox, O. H. E.**), "The CAMELS project: Cosmology and Astrophysics with MachinE Learning Simulations", *ApJ* **915**, 71 (2021) (arXiv).

84. Wang, Y., Zhao, G.-B., Zhao, C., **Philcox, O. H. E.**, *et al.*, "The clustering of the SDSS-IV extended Baryon Oscillation Spectroscopic Survey DR16 luminous red galaxy and emission line galaxy samples: cosmic distance and structure growth measurements using multiple tracers in configuration space", *MNRAS* **498**, 3470 – 3483 (2020) (arXiv).

85. *Chudaykin, A., Ivanov, M. M., **Philcox, O. H. E.**, Simonović, M., "Nonlinear perturbation theory extension of the Boltzmann code CLASS", *Phys. Rev. D* **102**, 063533 (2020) (arXiv).

***Other Works***

86. **Philcox, O. H. E.**, "An Introduction to the EFTofLSS", *Lecture Notes*.

## SELECTED RESEARCH TALKS

* = *Virtual Talk*

2026 Laboratoire d'Annecy de Physique Théorique, Annecy, France *Seminar*
CERN, Geneva, Switzerland *Seminar*
New Synergies in Multiprobe Cosmology, KITP, Santa Barbara, *Conference*

2025 41st Institut d'Astrophysique de Paris Symposium, *Conference* (Invited Talk)
Galileo Galilei Institute for Theoretical Physics, *Symposium* (Invited Plenary)
21st Rencontres de Vietnam, *Conference* (Invited Talk)
Understanding Cosmological Observations, Benasque, *Conference* (Invited Talk)
GR24 & Amaldi16, *Conference* (Invited Plenary)
Columbia University, *Physics Colloquium*
Cosmological Probes of New Physics, Notre Dame, *Conference* (Invited Discussion)
Brown University, *Theoretical Physics Seminar*
Princeton University, *Cosmology Seminar*
Particles vs. New Probes, Flatiron Institute, *Conference* (Invited Plenary)
Boston University, *High-Energy Physics Seminar*

2024 Essential Cosmology for the Next Generation IX, Mexico, *Winter School* (Invited Lecture Series)
21st Century Cosmology, Ashoka University, *Conference* (Invited Plenary)
Cosmology in the Adriatic, Split, *Conference*
University of Edinburgh, *Conference*
Lawrence Berkeley National Laboratory, *Workshop* (Invited Plenary)
University of Cambridge, *Astronomy Colloquium*
Royal Society, London, *Discussion Meeting* (Invited Talk)
Harvard University, *Cosmology Seminar*
Massachusetts Institute of Techology, *Cosmology Seminar*
58th Rencontres de Moriond, *Conference*
Stanford University, *Physics Colloquium*
Perimeter Institute for Theoretical Physics, *Colloquium*

2023 Taipei Institute of Astronomy & Astrophysics, *Conference* (Invited Talk)
Max-Planck-Institute for Nuclear Physics, Heidelberg, *Particle Physics Seminar*
ICTP Trieste, *Workshop*
University of Montreal, *Astrophysics Seminar*
*Parity-Violation from Home, *Conference*
Ohio State University, *Cosmology and Astro-Particle Physics Seminar*
*Early Universe / AliCPT Forum, University of Science and Technology of China, *Webinar*
Sexten Center for Astrophysics, *Workshop* (Invited Plenary)
Donostia International Physics Center, *Workshop* (Invited Talk)
University of Pennsylvania, *PDT Partners Retreat* (Invited Talk)
Yukawa Institute for Theoretical Physics, Kyoto, *Conference* (Invited Talk)
Kavli IPMU, Tokyo, *Astronomy Seminar*
Stony Brook, *Cosmology Seminar*
Cosmology on Safari, *Conference*
Johns Hopkins University, *Particle Physics Seminar*
University of Maryland, *Particle Physics Seminar*
*Copernicus Series, *Cosmology Webinar*
*University of Oxford, *Cosmology Seminar*

## OUTREACH TALKS

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|---|---|
| 2026 | *Astronomy on Tap*, San Francisco (Public Talk)<br>*Discover Our Universe*, Kavli Institute for Particle Astrophysics & Cosmology (Public Lecture) |
| 2025 | *Stanford University Undergraduate Physics Club* (Faculty Lunch Talk)<br>*Taste of Science*, New York (Public Lecture) |
| 2023 | *Into The Unknown* (Public Podcast) |

## MEDIA

1. "Scientists detect mysterious suppression in cosmic structure growth", *Phys.org*, 3 Jan. 2025.
2. "Universe's 'Cosmological Collider' Lands 3 Scientists $100,000 Physics Prize", *Live Science*, 14 Sep. 2023.
3. "What Happened Right After the Universe Began?", *Simons Foundation*, 30 Aug. 2023.
4. "The Cosmos as a Colloid", *Physics Magazine*, 14 Mar. 2023.
5. "Pinpoint Simulations Provide Perspective on Universe Structure", *IAS News* & *Phys.org*, 14 Mar. 2023.
6. "Spatial Pattens In Distribution of Galaxies", *Princeton News* & *ScienceDaily*, 14 Mar. 2023.
7. "Is the Universe Asymmetrical?", *Columbia News*, 27 Feb. 2023.
8. "Do We Live in a Mirror Universe?", *Into The Unknown Podcast*, 26 Jan. 2023.
9. "Asymmetry Detected in the Distribution of Galaxies", *Quanta*, 5 Dec. 2022.
10. "The Universe is Surprisingly Lopsided and We Don't Know Why", *New Scientist*, 18 Jun. 2022.

## PRIZES & GRANTS

| | |
|---|---|
| 2024 | New Horizons in Physics Prize, *Breakthrough Prize Foundation* |
| 2023 | Buchalter Cosmology Prize (First Prize) |
| 2022 | Simons Society of Fellows (Junior Fellowship<br>NHFP Einstein Fellowship, *declined*<br>LBL Chamberlain Fellowship, *declined*<br>Cambridge Kavli Fellowship, *declined* |
| 2018 | Herchel-Smith Scholarship, *Cambridge* → *Harvard*<br>Institute of Astronomy Prize, *Cambridge* |
| 2017 | Holgate Pollard Memorial Prize, *Cambridge* |

## PROFESSIONAL ACTIVITIES

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|---|---|---|
| **Referee** | MNRAS (2020−), JCAP (2020−), MPLA (2021−), Phys. Rev. Lett. (2022−)<br>Phys. Rev. D (2022−), ApJS (2022−), Phys. Dark Univ. (2023−) | |
| **Conferences** | Large-Scale Parity Violation (Taiwan 2023, SOC) | |
| **Review Panels** | NASA Astrophysics Theory Program (2023) | |
| **Mentor** | *Beatriz Tucci* | Stanford Postdoctoral Fellow (2025−) |
| **Advisor** | *Cooper Jacobus* | Stanford Graduate Student (2026, rotation) |
| | *Hanyue Wang* | Stanford Graduate Student (2026, rotation) |
| | *Sixiang Yang* | Stanford Graduate Student (2025, rotation) |
| | *Alessandro Russo* | Stanford Graduate Student (2025, rotation) |
| | *Anita Dunsmore* | Stanford Graduate Student (2025, rotation) |
| | *Sam Goldstein* | Columbia Graduate Student (2022−2025, various projects) |
| | *John Moynihan* | Columbia Undergraduate Student (2023, summer project) |
| | *Jess Boyland* | Simons-NSBP Undergraduate Scholar (2020−2021, summer project) |
| **Coadvisor** | *Luca Abu El-Haj* | Columbia Undergraduate Student (2025, summer project) |
| | *Gemma Zhang* | Harvard Graduate Student (2023−2025) |
| | *Kristen Surrao* | Columbia Graduate Student (2022−2023) |
| | *Gerrit Farren* | Cambridge Graduate Student (2020−2022) |

## MISCELLANEOUS

| | |
|---|---|
| **Computing Languages** | Python, C++, Julia, Mathematica, Cuda |
| **Codes Developed** | PolySpec, PolyBin3D, Encore, NPCFs.jl, Class-PT,<br>Spectra-Without-Windows, EffectiveHalos, Hipster, RascalC |
| **Teaching** | 6 years of online tutoring (high-school to Masters level)<br>Teaching assistant for Princeton introductory astronomy class (AST203)<br>TEFL qualification in English teaching |
| **Other** | DipABRSM in Music Performance (Distinction) |

## REFERENCES


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